

6.8 Quiz: Polynomials — Markscheme (b)

1. [Maximum mark: 7]

Let $f(x) = x^2 + 6x - 16$.

(a) Write $f(x)$ in factored form $(x - a)(x - b)$. [1]

$$f(x) = (x - 2)(x - (-8)) = (x - 2)(x + 8) \quad \text{(M1)(A1)}$$

Accept $(x + 8)(x - 2)$.

(b) Write down the x -intercepts. [1]

$$x = 2 \text{ and } x = -8 \quad \text{(A1)}$$

(c) Write down the y -intercept. [1]

$$f(0) = -16, \text{ so the } y\text{-intercept is } (0, -16). \quad \text{(A1)}$$

(d) Write $f(x)$ in vertex form $(x - h)^2 + k$. [1]

$$f(x) = (x + 3)^2 - 25 \quad \text{(A1)}$$

i.e. $h = -3, k = -25$.

(e) Find the coordinates of the vertex. [2]

$$\text{Axis of symmetry: } x = \frac{-8 + 2}{2} = -3 \quad \text{(A1)}$$

$$f(-3) = 9 - 18 - 16 = -25$$

$$\text{Vertex: } (-3, -25) \quad \text{(A1)}$$

2. [Maximum mark: 7]

Consider the function $g(x) = -(x - 5)^2 + 4$.

(a) Write down the coordinates of the vertex. [1]

$$\text{Vertex: } (5, 4) \quad \text{(A1)}$$

(b) Write down the equation of the axis of symmetry. [1]

$$x = 5 \quad \text{(A1)}$$

(c) Show that $g(x) = -x^2 + 10x - 21$. [2]

$$g(x) = -(x - 5)^2 + 4 = -(x^2 - 10x + 25) + 4 \quad \text{(M1)}$$

$$= -x^2 + 10x - 25 + 4 = -x^2 + 10x - 21 \quad \text{(A1)(AG)}$$

(d) Solve $g(x) = 0$ to find the x -intercepts. [3]

$$-x^2 + 10x - 21 = 0 \quad \textbf{(M1)}$$

$$x^2 - 10x + 21 = 0$$

$$(x - 3)(x - 7) = 0 \quad \textbf{(A1)}$$

$$x = 3 \text{ or } x = 7 \quad \textbf{(A1)}$$

3. [Maximum mark: 7]

The line $y = 4x$ and the parabola $y = x^2 - 5$.

(a) Show that the x -coordinates of intersection satisfy $x^2 - 4x - 5 = 0$. [2]

At intersection: $x^2 - 5 = 4x$ (M1)

$x^2 - 4x - 5 = 0$ (A1)(AG)

(b) Show that $x^2 - 4x - 5 = 0$ has two distinct real roots. [1]

$\Delta = (-4)^2 - 4(1)(-5) = 16 + 20 = 36 > 0$ (A1)

Since $\Delta > 0$, there are two distinct real roots.

(c) Factorise $x^2 - 4x - 5$. [1]

$(x - 5)(x + 1)$ (A1)

(d) Find the x -coordinates of the points of intersection. [1]

$x = 5$ and $x = -1$ (A1)

(e) Find the coordinates of both points of intersection. [2]

$x = 5: y = 4(5) = 20 \implies (5, 20)$ (A1)

$x = -1: y = 4(-1) = -4 \implies (-1, -4)$ (A1)

4. [Maximum mark: 7]

The equation $2x^2 - kx + 32 = 0$ has two equal roots.

(a) Write down the discriminant in terms of k . [1]

$\Delta = (-k)^2 - 4(2)(32) = k^2 - 256$ (A1)

(b) Find the values of k . [2]

Equal roots: $\Delta = 0$

$k^2 - 256 = 0 \implies k^2 = 256$ (M1)

$k = \pm 16$ (A1)

(c) For $k = 16$, find the repeated root. [2]

$2x^2 - 16x + 32 = 0 \implies x^2 - 8x + 16 = 0$ (M1)

$(x - 4)^2 = 0 \implies x = 4$ (A1)

(d) State the values of k for which the equation has no real roots. [2]

$$\Delta < 0 \implies k^2 - 256 < 0 \implies k^2 < 256 \quad \textbf{(M1)}$$

$$-16 < k < 16 \quad \textbf{(A1)}$$